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5) Hallar todos los $x \in \mathbb{R}$:

$$|2x-6| \geq x^2 - 5x + 6$$

1) Si $2x-6 \geq 0$

$$x \geq 3$$

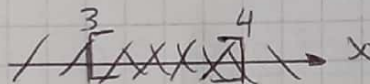
$$2x-6 \geq x^2 - 5x + 6$$

$$0 \geq x^2 - 7x + 12$$

$$\frac{7 \pm \sqrt{49 - 4 \cdot 12}}{2} = \frac{7 \pm 1}{2} = x_1 = 4$$

$$x_2 = 3$$

$$f(2) = 2^2 - 7 \cdot 2 + 12 = 2 +$$



2) Si $2x-6 \leq 0$

$$x \leq 3$$

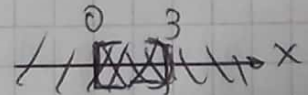
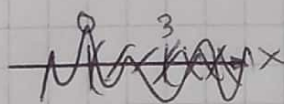
$$-2x+6 \geq x^2 - 5x + 6$$

$$0 \geq x^2 - 3x$$

$$0 \geq x(x-3)$$

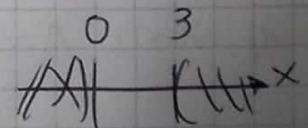
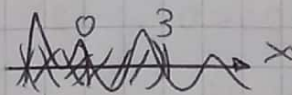
$$x \geq 0 \wedge x-3 \leq 0$$

$$x \leq 3$$



$$x < 0 \wedge x-3 > 0$$

$$x > 3$$



$$S = [3, 4] \cup [0, 3]$$

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